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Analyzing Publicly Reported Waiting Times Data in Portuguese Health Service

Bruno Oliveira[a]*, Daniel Teixeira[a], Diogo Silva[a], Vânia Oliveira[a], João Oliveira[a], Luís Soares[a]

[a]*CIICESI, Escola Superior de Tecnologia e Gestão, Instituto Politécnico do Porto, Portugal*

Abstract

This paper presents the development and implementation of a Business Intelligence (BI) system designed to analyze publicly reported waiting time data from the Portuguese National Health Service (SNS). Data were collected through a public Application Programming Interface (API), including near real-time emergency service updates (every 15 minutes), enabling more detailed temporal analyses. The system integrates and transforms raw data into structured analytical outputs, offering insights into waiting time disparities across institutions, specialties, and regions. It also evaluates data quality issues such as refresh frequency and inconsistencies. Our results highlight systemic inefficiencies and suggest opportunities for improving public health transparency and operational planning.

Keywords: Waiting Times; Portuguese Health Service; Healthcare Data Quality; Data Analytics; Emergency Services Data

1. Introduction

The availability and accessibility of public data have become increasingly important for promoting transparency and accountability in health services [1]. In the healthcare domain, timely and accurate data are essential not only for evaluating service performance [2], but also for informing policy decisions and improving patient outcomes [3]. In

* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000 .
E-mail address: bmo@estg.ipp.pt

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Portugal, the National Health Service (SNS) provides public access to healthcare waiting time data through an open API, enabling external stakeholders—researchers, policymakers, and citizens—to monitor and analyze the performance of healthcare services. Despite the value of this initiative, the data provided by the SNS present several limitations, including inconsistent update frequencies, potential discrepancies in the information reported by institutions, and weak alignment with the actual operational status of the healthcare system. These challenges raise questions about the reliability and usability of such data for analytical purposes.

This paper presents the development of a Business Intelligence (BI) system designed to address these limitations by systematically collecting, transforming, and analyzing data retrieved from the SNS public API. The system integrates multiple endpoints—covering emergency services, consultations, and surgeries—into a unified data warehouse, enabling multidimensional analyses across specialties, institutions, and regions. In addition to examining temporal trends and regional disparities, the system includes mechanisms for data profiling and monitoring data freshness, thus offering a more structured and critical view of the publicly available information. The study addresses key questions concerning the reliability and analytical potential of public API data: What patterns and trends can be identified in waiting times across services and regions? To what extent do the data reflect consistent and up-to-date reporting practices? And how can such data support evidence-based service evaluation and performance monitoring?

By providing a case study focused on the Portuguese health system, this work contributes to the broader discussion on the role of public data in supporting transparency, healthcare planning, and informed decision-making. These aspects are particularly relevant for fostering public trust and ensuring accountability in public service delivery. The remainder of this paper is organized as follows. Section 2 reviews related work and describes the methodology used in the analysis. Section 3 presents the system architecture, data sources, and analytical results. Section 4 discusses the implications of the findings, particularly concerning data quality and operational monitoring. Finally, Section 5 concludes the paper and outlines directions for future development.

2. Related Work

Publicly accessible healthcare datasets are increasingly recognized as valuable resources for supporting evidence-based policy-making and system evaluation. By offering reliable and transparent data, such resources enable the identification of critical health trends, the assessment of policy impacts, and the development of targeted interventions [1]. In this context, [4] propose an analytics framework based on open-source tools to explore public health data, combining it with demographic and epidemiological information to enrich analytical insights.

However, the use of public API data for healthcare service evaluation, particularly concerning waiting times, presents several challenges. As shown in [5], inconsistencies in how countries measure waiting times—including differences in parameters, service definitions, and reporting granularity—complicate international comparisons. Their analysis across 15 countries revealed that while many track national waiting time statistics and enforce care guarantees, the lack of standardization limits the comparability and applicability of these datasets across borders. This highlights the need for greater alignment in data collection protocols and definitions. In addition to the policy and data governance aspects, the technical foundations of healthcare analytics have been extensively studied. Data warehouses play a central role in consolidating heterogeneous data sources, enabling timely access to comprehensive and structured healthcare information. As noted in [6], the integration of clinical and operational data in hospitals enhances evidence-based medicine by supporting more accurate diagnoses, personalized treatment plans, and improved clinical decision-making. Beyond clinical care, data warehousing also supports strategic hospital management by optimizing resource allocation, informing negotiations in managed care, and supporting pathway implementation. When coupled with data mining techniques, these systems can aid in identifying therapy options tailored to patient profiles.

Recent developments further demonstrate the applicability of data integration in large-scale health crises. For example, [7] implemented a multicenter data warehouse based on Electronic Health Records (EHRs) to track the progression of COVID-19 and guide treatment strategies across multiple hospitals. Similarly, domain-specific data warehouses—such as those designed for lymphoma diagnosis and treatment planning—have proven effective in supporting personalized medicine [8]. While the studies above primarily focus on clinical and hospital-based data systems, our work contributes a complementary perspective by analyzing open government data from the Portuguese National Health Service (SNS). This dataset is made available through a unified API platform, governed by consistent publication guidelines. We aim to explore the mechanisms of access, assess the structure and completeness of the available data, and evaluate its suitability for analytical use. Although this study does not directly measure the accuracy of the data in representing real-world events, it emphasizes the analytical potential of these public resources.

More specifically, this paper presents a Business Intelligence (BI) system designed to extract, transform, and analyze SNS waiting time data, to generate detailed and actionable insights for healthcare stakeholders. By equipping users with tools to visualize regional disparities, monitor institutional reporting behavior, and identify service bottlenecks, we illustrate how open data—despite its imperfections—can play a strategic role in improving healthcare system transparency, efficiency, and responsiveness.

3. Hospitals Waiting Times

The primary data source for this project is the Average Waiting Times platform provided by the Portuguese National Health Service[†]. This platform offers detailed information on waiting times at Portuguese hospitals, covering various services including emergency care, outpatient consultations, and surgeries. Although it provides a comprehensive perspective for analyzing healthcare service performance, it is important to consider the characteristics and limitations of this data source. These include inconsistencies and variations in data accuracy and reliability, which may arise from challenges in data collection, processing, and updating. This study explores the available data and examines its implications for evaluating service quality and supporting informed decision-making.

3.1. Data Collection

The data for this project were obtained through requests to APIs provided by the Average Waiting Times platform of the Portuguese National Health Service (SNS). These requests enabled the collection of detailed information on waiting times across various hospital services and contexts.

```json
{
    "LastUpdate": "2025-02-16T11:55:53.543",
     (...)
    "Queue": null,
    "ScaleType": 0,
    "Red": {
        "Time": 0,
        "Length": 0
    },
    "Orange": {
        "Time": 326,
        "Length": 0
    },
    "Yellow": {
        "Time": 330,
        "Length": 1
    },
    "Green": {
        "Time": 965,
        "Length": 0
    },
    "Blue": {
        "Time": 0,
        "Length": 0
    }
}
```

Figure 1. Example JSON Data Structure

[†] https://tempos.min-saude.pt/\#/instituicoes

According to the technical specification for emergency waiting times[‡], the API provides approximate waiting times for emergency services based on data reported by each institution. Hospitals are responsible for calculating and submitting these values. Each institution must report waiting times by hospital unit, triage counter (identified by hospital location), and priority level following the Manchester Triage System. The waiting lines are organized by clinical priority and can be further subdivided by specialty, such as General Care, Surgery, or Orthopedics. Institutions must provide unique identifiers and descriptions for each triage counter and queue, and report the number of patients waiting for consultation by priority. If multiple queues exist for the same priority level, the queue with the longest wait time should be displayed. Updates must be submitted every 15 minutes, and a timestamp is shown to indicate the last update. If no updates are received within two hours, the system flags the institution as not currently sharing waiting times. Figure 1 shows an example of JSON data representing average waiting times for a specific institution.

Data collection was performed using various strategies. For emergency care, real-time collection enabled the capture of dynamic variations in waiting times. For consultations and surgeries, data were extracted between August and October, with the API providing data segmented by monthly intervals. The types of services offered vary across hospitals, particularly in more remote areas, complicating comparative analysis. Another relevant factor is the potential influence of the initial screening conducted by the “Saúde 24” service. "Saúde 24" is a national health service in Portugal that provides telemedicine and remote healthcare assistance. In cases of urgent medical issues, it may redirect individuals to emergency rooms, specialized clinics, or general practitioners. This redirection can affect both the distribution and the recorded waiting times.

There are substantial differences between hospitals in how they report waiting times. Some hospitals only provide data for general emergency categories, such as General Emergency and Pediatric Emergency, while others include internal subdivisions, such as Surgery, Orthopedics, and Ophthalmology within the same category. These differences were taken into account during the analysis to ensure that the system could categorise similar services uniformly. This allows for detailed analyses by emergency type and service level, as well as the identification of data gaps that may affect the interpretation of results. A comprehensive understanding of the data source is essential for ensuring the quality and reliability of the analyses, and for adapting the system to the specific characteristics of hospital services in Portugal.

Additionally, we collected data from a prominent Portuguese private healthcare provider to broaden the scope of the analysis and evaluate the feasibility of future integration of private sector data. However, to our knowledge, there are no public APIs offering standardized access to private hospital data. In addition, private hospitals are not subject to the same reporting regulations as public institutions, resulting in differences in data structure and update frequency. For example, while public hospitals are expected to update their data every 15 minutes, some private hospitals update every 5 minutes, and in other cases, the update frequency is unclear. Due to these inconsistencies, direct comparisons between public and private hospitals are not currently feasible. Nevertheless, we included sample data from private hospitals to prepare the system for potential future integration, ensuring adaptability while recognizing the current limitations in comparability.

3.2. Analysing Data

After data collection and characterization, a data profiling process was conducted to assess the quality, consistency, and structure of the available data. The main findings from this stage revealed missing values in fields such as average waiting times for certain specialties and institutions, which were labeled as "Data Unavailable." Inconsistencies were observed, including duplicate entries for emergency types and specialties with slight variations, which were resolved through normalization. The data showed high variability in average waiting times, especially in specialties like Orthopedics and Ophthalmology. Triage types and priority levels generally followed expected standards. However, the calculation of `TimeComplianceDelay` might be limited by missing intermediate records due to periods of API unavailability. Figure 2 presents the schema of the dimensional model developed to support the analysis. The model consists of fact and dimension tables, each serving a specific purpose in structuring and analyzing healthcare data.

[‡] https://www.spms.min-saude.pt/wp-content/uploads/2024/10/ET-RSE-WebAPI_v1.6.1.pdf

The fact tables store numerical and aggregated metrics, such as the number of attendances, average waiting times, and the percentage of cases attended within the expected timeframe. The main fact tables include `fact_emergencies`, which records average and total waiting times for emergency services, segmented by priority level and hospital, and `fact services`, which stores similar data for consultations and surgeries, categorized by specialty and hospital.

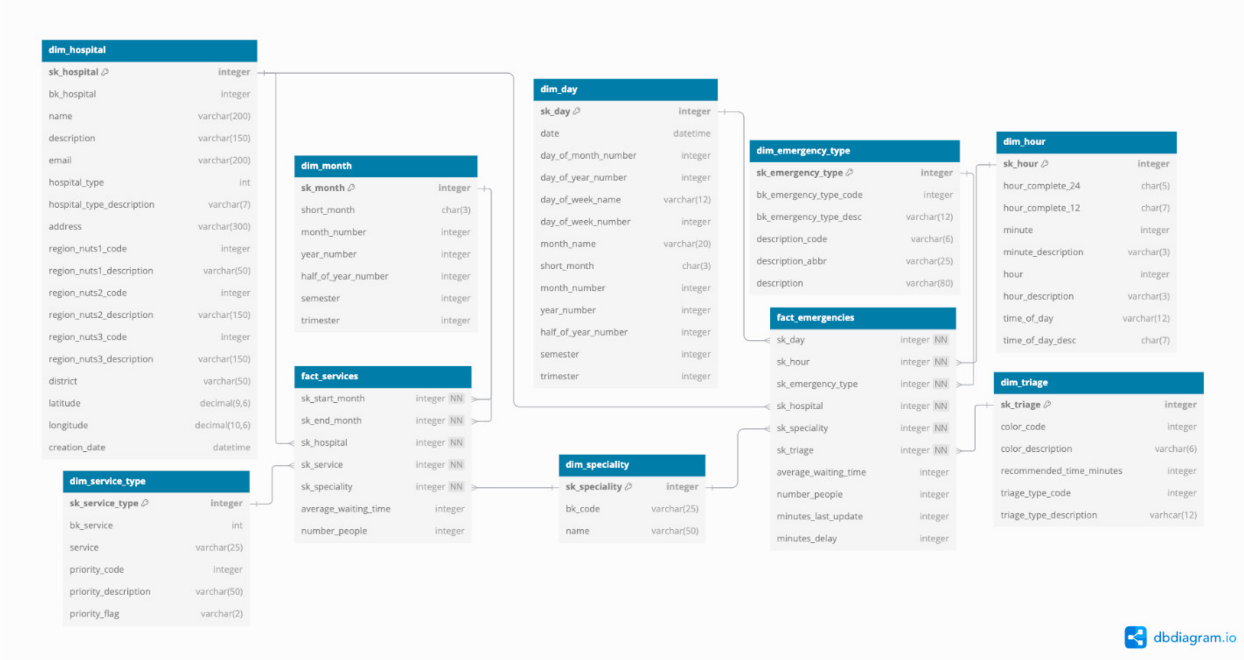

Figure 2. Data Warehouse Schema

The dimensional model includes several dimension and fact tables designed to support detailed and flexible analysis of healthcare services. The dimension tables provide contextual information for the fact tables. The `dim_hospital` table contains attributes for each hospital, including name, type (public or private), geographical location, and administrative region classification (NUTS1, NUTS2, and NUTS3). The `dim_emergency_type` table categorizes emergency case types, including codes, abbreviations, and full descriptions. The `dim_triage` table details triage systems, associating types with corresponding color codes and recommended waiting times. The `dim_speciality` table classifies medical specialties. The `dim_service_type` table categorizes services (consultations and surgeries) by priority level.

Time-related dimensions include `dim_day`, which offers granularity at the day level with attributes such as day of the month, day of the year, day of the week, month, trimester, and semester. The `dim_hour` table adds hourly granularity with fields for 24-hour and 12-hour formats, minute intervals, and time-of-day classifications (e.g., morning, afternoon, evening). The `dim_month` table aggregates data at the monthly level, with information on month number, year, semester, and trimester, which is important for long-term trend analysis. The `fact_emergencies` table focuses on emergency room emergencies and includes the following metrics:

- `average_waiting_time`: Average wait time (in minutes) before patients receive emergency care.
- `number_people`: Total number of individuals waiting for emergency services.
- `minutes_delay`: Delay relative to the recommended waiting times by triage priority, indicating compliance.
- `minutes_last_update`: Time (in minutes) since the last data update, used to track data freshness.

This table is linked to the `dim_day`, `dim_hour`, `dim_hospital`, `dim_emergency_type`, `dim_speciality`, and `dim_triage` dimension tables. Similarly, the \texttt{fact_services} table captures performance indicators for consultations and surgeries. It includes:

- `average_waiting_time`: Average waiting time for consultations and surgical procedures.
- `number_people`: Number of individuals in queue for a specific service, relevant for capacity planning.

This table references `dim_hospital`, `dim_speciality`, `dim_service_type`, and the time-related dimension `dim_month`, for both the start and end periods of service data collection. Together, the dimension and fact tables provide a comprehensive view of healthcare service efficiency, supporting data-driven decision-making and ongoing performance monitoring.

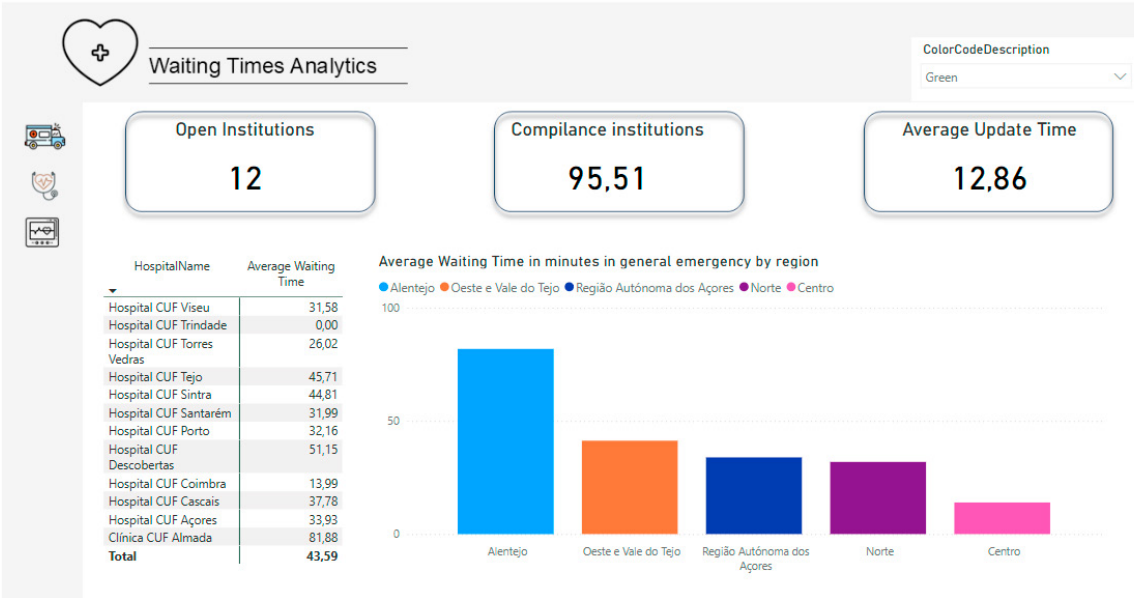

Figure 3. Power BI Report Example

3.3. Implementation

The implementation of the Business Intelligence system was supported by a carefully selected set of tools, each chosen to address different stages of the project, from data storage to results visualization.

MongoDB was used as the primary database for storing and processing the collected data. Its flexibility in handling semi-structured data, such as JSON files, along with its robust aggregation framework, made it suitable for processing, cleaning, and transforming diverse data formats before loading them into the data warehouse. The data warehouse was implemented using a relational database engine (PostgreSQL), designed to support efficient querying and analysis through a dimensional model. The ETL process used MongoDB's aggregation framework for data processing and quality enhancement in the staging area. Final data loading into the relational database was performed using SQL Server Integration Services (SSIS), enabling structured and optimized data transfer.For data visualization and analysis, Power BI was used to create interactive dashboards integrated with the dimensional model. Within Power BI, several enhancements were implemented, including the creation of hierarchies to support drill-down analysis, calculated columns for refined data manipulation, and custom metrics to deliver more meaningful insights into hospital waiting times. These elements contributed to a dynamic and intuitive reporting environment, enabling effective data-driven decision-making.

4. Discussion and Analysis

The analyses performed with the developed BI system were organized across three core domains: emergency services, outpatient consultations, and surgical procedures. Complementary analyses were also conducted at the regional level and across short time intervals to capture early indicators of seasonal effects and temporal variation in service demand. In the context of emergency services, average waiting times by triage priority revealed marked regional disparities. The regions of Alentejo and the Azores consistently exhibited the longest waiting times across all triage levels. Furthermore, temporal patterns indicated that delays tend to peak during night shifts, which may reflect staffing constraints during off-peak hours. A closer examination of triage categories showed that patients in the "green" and "orange" classifications—representing non-urgent and moderately urgent cases—experienced the most substantial delays, especially during periods of high demand. These findings suggest that existing resource allocation strategies may be insufficient for maintaining responsiveness to lower-priority cases, potentially undermining the overall efficiency and equity of emergency care delivery.

In the consultation domain, the analysis uncovered significant variability in waiting times across specialties and institutions. Specialties such as Ophthalmology and Orthopedics consistently showed the longest average delays, often surpassing 100 days in certain hospitals. Even in the case of urgent consultations, the average waiting time was close to 30 days, revealing only partial compliance with service targets. By disaggregating consultation delays by specialty, priority level, and institution, clear contrasts emerged that point to both systemic bottlenecks and localized inefficiencies. For example, while some hospitals reported relatively short delays in specialties like Internal Medicine, others showed pronounced delays in areas such as Anesthesiology and General Surgery, emphasizing the need for more granular and targeted planning. The surgical waiting time analysis confirmed that Orthopedics and General Surgery are among the most heavily burdened specialties. Once again, Alentejo and the Azores stood out for their significantly longer surgical wait times, underscoring persistent capacity limitations in peripheral regions. These results collectively reinforce the hypothesis that regional disparities in infrastructure and staffing levels may be driving unequal access to timely surgical care. Addressing these disparities will likely require coordinated interventions focused on decentralization, referral optimization, and investment in underserved areas. Although the available dataset did not span a full calendar year—limiting the feasibility of a full seasonal analysis—short-term patterns were nonetheless observed. Temporal fluctuations in service demand and delay behavior within the observed period highlight the importance of capturing finer temporal granularity in future datasets. Expanding data coverage over time could allow more robust detection of seasonal surges and longer-term performance trends. In parallel with performance indicators, the system was used to monitor institutional compliance with the mandated 15-minute update interval for emergency service data. The first phase of this analysis focused on measuring the average delay between reported updates. Most institutions exhibited average delays ranging from 5 to 10 minutes, suggesting general but imperfect adherence to the expected interval. An hourly analysis of delay percentages revealed a peak around 10 AM, followed by relatively stable patterns throughout the remainder of the day. This concentration of delays in the morning may reflect operational constraints, system maintenance schedules, or workload accumulation at the start of hospital shifts. Daily update delay patterns further revealed that the most significant delays occurred during the initial days of the observation period. Over time, delay rates declined slightly, which may indicate the impact of system stabilization, human adaptation to the reporting process, or underlying operational improvements. To better understand institutional behavior, we examined the percentage of updates that exceeded the 15-minute threshold across all hospitals. The results showed that these delays were not concentrated in a handful of institutions but were rather widespread. Moreover, the persistence of delays across the six weeks suggests that the issue may reflect structural rather than circumstantial causes. These insights provide a valuable foundation for developing recommendations aimed at improving reporting compliance and, by extension, the overall reliability of the data.

The findings presented highlight disparities and possible inefficiencies in the Portuguese health system. Once again, Alentejo and the Azores emerge as regions with consistent underperformance in terms of both service responsiveness and data reporting. Moreover, the preliminary evidence of seasonal variability and systemic update delays underscores the importance of proactive resource planning—particularly in human resources, facilities, and digital infrastructure. The integration of predictive analytics into hospital management systems could further enhance the ability to anticipate and mitigate future surges in demand. These insights offer practical value for hospital administrators, policy-makers, and oversight agencies. By leveraging data-driven approaches to monitor real-time performance, institutions can make more informed decisions about resource allocation, prioritize critical specialties, and promote more equitable access to healthcare. Figure 3 presents an example of a Power BI dashboard generated from the system, illustrating key indicators segmented across emergencies, consultations, and surgeries. Finally, it is important to recognize the limitations of the current study. The analyses are based on a limited observation period and do not account for unreported services or private healthcare dynamics. Therefore, the results presented here should not be interpreted as definitive system-wide conclusions. Rather, they should be viewed as a demonstration of the analytical capabilities unlocked by integrating public data into a structured BI environment. With continued data collection and methodological refinement, this system can evolve into a robust tool for ongoing healthcare monitoring and service improvement.

5. Conclusions and Future Work

By integrating publicly available data from the SNS platform into a dimensional data warehouse and building interactive dashboards in Power BI, the system provided a structured and accessible means to evaluate service performance in three critical domains: emergency care, medical consultations, and surgical procedures. The analyses uncovered significant disparities between institutions and regions, generating valuable insights to support strategic decision-making and healthcare planning. The system also enabled the identification of operational challenges, including recurring delays in data updates and inconsistencies in service categorization across institutions. These issues, although outside the control of the system itself, represent important obstacles to achieving reliable and timely health monitoring. By systematically quantifying update compliance and revealing areas of delayed reporting, the platform contributes to promoting greater accountability and transparency in healthcare delivery. Despite meeting its initial objectives, the project faced some limitations. The available dataset covered only a short observational period, restricting the possibility of performing full seasonal analyses or identifying long-term trends. Differences in the structure and accessibility of data between public and private providers also posed challenges for integration, often requiring extensive manual standardization. Moreover, the analysis focused exclusively on operational metrics—such as waiting times and data freshness—without incorporating complementary indicators like patient satisfaction, clinical outcomes, or financial efficiency, which would be necessary for a more comprehensive performance assessment.

Future work will focus on expanding the system's analytical scope and robustness. Incorporating predictive modeling techniques based on machine learning could support proactive resource planning by forecasting demand surges and identifying emerging bottlenecks before they become critical. This aim not only to improve the system's technical and analytical capabilities but also to increase its relevance for policymakers, hospital administrators, and other stakeholders. By continuing to leverage open data and transparent methodologies, the system has the potential to contribute meaningfully to efforts to improve efficiency, equity, and patient outcomes across the Portuguese healthcare system.

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